

Report

Janani Sri Sree Ganesh

May 2025

1 Automated Loan Approvals – An Overview

In this project, I investigate automated loan approval models and assess their alignment with the EU AI Act. An automated loan approval system uses algorithmic screening to evaluate applications based on a wide range of borrower and loan characteristics. These models estimate an applicant's likelihood of repayment and determine whether a loan should be approved or rejected. It depicts the frameworks of screening and adverse selection by automatically filtering applicants and also prevents the pooling of high- and low-risk borrowers. As part of the PhD Coding Challenge, I built two algorithms with six variants to automate this decision-making process.

2 Characteristics and Limitations of the Dataset

I worked with a dataset (`loan_dataset.csv`) consisting solely of approved loan applications from January 2020 to December 2022. It includes variables on borrower characteristics, loan details, and demographic factors. However, a major limitation I encountered is that the data set does not contain rejected loan applications. This creates a selection bias as the data only reflects behavior conditional on approval. Therefore, it is challenging to understand the structural relationship between applicant attributes and approval decisions. To address this issue, counterfactual data such as rejected applications would be necessary. Additionally, I recognized that the dataset could perhaps inherit historical biases embedded in past human decision-making as the datasets inputs were hand collected by loan officers. If previous approvals made by the loan officers reflected discriminatory heuristics, the model could reinforce these patterns. The dataset is also limited in time coverage (Jan 2020 to Dec 2022). This spans a period that was affected by macroeconomic shocks and uncertainty which may make generalizing the results hard.

2.1 Sanity Checks

As part of my data validation process, I conducted several sanity checks and identified a notable anomaly. 5,694 observations reported impossible work experience values, i.e., borrowers claiming more years of work experience than possible given their age and Belgium's legal working age of 15. For instance, customer id 31836 who applied when he was 50 years old had already had 40 years of work experience at his current company.

2.2 Patterns and Key Relationships

Through exploratory data analysis, I uncovered a few meaningful patterns. The overall repayment rate in the dataset is 62.43%, while the default rate stands at 37.56%. This also indicates a class

imbalance. I observed that zip code prefix 10 has the highest default rate (41%) and the lowest number of approved loans. Without data on application volume or exogenous shocks affecting this region, it is difficult to determine whether this pattern reflects discrimination, low application rates, or genuine borrower risk.

The total unpaid amounts from defaulters were relatively uniform across demographic groups such as age, education, zip area, and home ownership. However, borrowers who successfully repaid their loans tended to have higher median savings than those who defaulted. Interestingly, the proportion of borrowers with co-signers was nearly identical across both groups. Furthermore, 61% of those who repaid were existing customers, compared to 58% among defaulters.

Finally, I categorized credit scores into five bins: Poor (0–580), Fair (580–670), Good (670–740), Very Good (740–800), and Excellent (800+). The results were very telling: 90% of borrowers in the 'Poor' category defaulted, while 50% in the 'Fair' category did so. This result could indicate the cost efforts (to increase or maintain their credit scores) by high type borrowers that leads to a separating equilibrium between high- and low-type borrowers.

3 Modeling approach and results

The loan approval model is a binary classification problem. I evaluate two algorithms and six variants, in total. A plain logistic-regression model with class weighting is the baseline model, and a linear support-vector classifier is its margin-based comparison. Both algorithms are first run on the raw, unscaled features, then rerun inside a scaling pipeline. Finally, I endow each classifier with an L1 penalty so that it can discard non-informative dummies and produce a sparser, more interpretable coefficient vector. Evaluation is based on ROC AUC score. The unscaled logistic model starts with an ROC AUC of 0.68, while the unscaled linear SVC lags slightly behind. Once RobustScaler is inserted, both models leap into the mid-0.83 range and get better with the L1 penalty. Since, the performance scores are very similar across the final four variants, I chose the simplest logistic regression algorithm as my final model.

Model	Pre-processing / Variant	ROC AUC	PR AUC
Logistic Regression	Unscaled	0.682	–
Linear SVC	Unscaled	0.619	–
Logistic Regression	Scaled (RobustScaler)	0.835	–
Linear SVC	Scaled (RobustScaler)	0.835	–
Logistic Regression + L1	Scaled (RobustScaler)	0.837	0.821
Linear SVC + L1	Scaled (RobustScaler)	0.836	–

Table 1: Performance metrics for candidate models.

4 Alignment with EU AI Act Principles

In line with the principles outlined in the EU AI Act, my loan approval model covers most requirements including transparency, fairness, and human oversight. First, I believe that the dataset provides sufficient information to screen and separate low-risk and high-risk borrowers using observable signals or the datasets features. However, since the data includes only approved loans, it

suffers from selection bias. To uncover the structural relationship between borrower characteristics and loan approval outcomes, access to counterfactuals would be necessary.

Regarding human oversight, the model incorporates a human-in-the-loop mechanism. Specifically, if the logistic regression model predicts a probability of loan repayment greater than 0.8 or less than 0.2, the application is flagged for manual review. This is designed in such a way that the edge cases are not left solely to algorithmic judgment.

Proxy variable / group	Δ Selection Rate
HS Diploma	0.045
Bachelors	0.002
Masters	0.040
PhD	0.039
Mortgage	0.019
Own	0.025
Rent	0.000
In Person	0.023
Online	0.028
Phone	0.020
Business	0.030
Car	0.010
Home Improvement	0.031
Other	0.015

Table 2: Selection-rate disparity (largest absolute difference between sub-groups) for each proxy variable in the final plain logistic-regression model, evaluated on the July–December2022 test set.

For fairness and non-discrimination, I test the model using demographic features as proxies for protected attributes. I calculate differences in selection rates across one-hot encoded groups, based on whether the model classifies the test set’s applicants into “repaid” or “default” categories. Across all fairness checks, these differences are negligible, indicating that the model does not systematically disadvantage any specific group. Nonetheless, I recognize that fairness audits must go beyond surface-level statistical checks. Even if the current model passes these tests, future improvements or more powerful algorithms must be regularly monitored so they do not learn and reinforce latent discriminatory patterns.

5 Recommendations

The key recommendations that I would suggest are as follows. One, the hand collected data which has self-reported values of the borrowers should be consistent and clear. As mentioned earlier, the work experience observations pose doubt about the quality of self-reported data. Next, regarding the dataset itself, it will be helpful to obtain the rejected loans dataset with continuous rolling updates. Where GDPR allows, if we store protected attribute fields, it will make fairness audits more robust. Regarding the model threshold, certain data segments like “poor” credit score bin and Zip code prefix 10 are sparse, so until there are more observations auto-decisions can be removed for these segments alone.